

## **Business Rules: Lighting & Chandelier Shop**

1. The shop sells four types of products: Chandeliers, Wall Clocks, Wall Lights, and Lamps (Table, Floor, Bedside).
2. Each product has a unique code, a name, a type, a purchase price, a selling price, and a stock quantity.
3. The selling price of every product is set at a fixed markup of 25% above its purchase price.
4. A product cannot be sold if its stock quantity is zero, and stock quantity can never be negative.
5. The shop is managed by a single manager who handles all operations, identified by an ID(phone number) and a name.
6. The shop is identified by a name and has a fixed address. This information appears on every purchase invoice issued.
7. The shop deals with exactly three suppliers: Al-Rayan, Maa Al-Dahab, and Al-Muttahida.
8. Each supplier is identified by an ID(phone number) and has a name, and an address.
9. Every Sunday, the manager reviews stock levels and contacts one or more suppliers by phone to place orders for needed products.
10. Each purchase order results in a purchase invoice that belongs to one and only one supplier, and is issued by the one and only manager.
11. A purchase invoice must contain at least one product line, and each line records the product code, name, type, quantity ordered, and unit price.
12. Each purchase invoice records the date of delivery, the total price of the order, the manager's ID(phone number) and name, the shop's name and address, and the supplier's name, phone number, and address.
13. A product can appear in multiple purchase invoices across different suppliers, and each supplier can provide many products.
14. Payment to suppliers must be made immediately at the time of delivery, in cash or by card; deferred payments not allowed.
15. Each customer is uniquely identified by their phone number.

16. The manager contacts customers by phone after purchase to collect feedback.
17. A customer can place one or more sales invoices over time, but each sales invoice belongs to one and only one customer.
18. A sales invoice must contain at least one product line, and each line records the product code, name, type, quantity, and unit price.
19. Each sales invoice records the invoice date, the total price after discount, the manager's ID(phone number) and name, the shop's name and address and the customer's phone number.
20. A product can appear in multiple sales invoices, and each sales invoice can include multiple products.
21. Payment from customers must be made immediately at the time of purchase, in cash or by card; no installments or deferred payments are allowed.
22. A customer whose first recorded invoice in the system is 6 or more months old is considered a loyal customer and automatically receives a 10% discount on their invoice.
23. A customer whose single invoice total reaches 4,000 EGP or more before discount is considered a bulk buyer and receives a 15% discount on that invoice.
24. If a customer qualifies for both discounts, only the higher discount of 15% is applied; discounts are never stacked.
25. A customer may return a product within 14 days of the original sales invoice date, provided the product is in undamaged condition.
26. Each sales invoice can have at most one return, and each return is linked to one and only one sales invoice.
27. Upon a successful return, the original sales invoice is marked as cancelled in the system, and the returned product quantity is restored to stock.